

LOUIS RIVOIRE, Ph.D.

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lrivoire.github.io

EDUCATION

Ph.D., Atmospheric Science Colorado State University, <i>Fort Collins, CO</i>	2020
M.S., Ocean, Atmosphere, Climate, and Remote Sensing Sorbonne University, <i>Paris, France</i>	2015
Diplôme de l'Ecole Normale Supérieure Advanced interdisciplinary research training, <i>Paris, France</i>	2015
B.S., Earth Sciences Ecole Normale Supérieure, <i>Paris, France</i>	2013
Classe Préparatoire aux Grandes Ecoles Intensive program in mathematics, physics, life sciences, and humanities, <i>Lyon, France</i>	2010-2012

PROFESSIONAL EXPERIENCE

Visiting Scientist Weizmann Institute of Science, <i>Rehovot, Israel</i> Department of Earth and Planetary Sciences	since Feb 2026
Postdoctoral Associate Massachusetts Institute of Technology, <i>Cambridge, MA</i> Department of Earth, Atmospheric, and Planetary Sciences	Jan 2024 - Dec 2025
Scientific Consultant NASA Jet Propulsion Laboratory, Caltech, <i>Pasadena, CA</i> Space mission concept design	Dec 2021 - Nov 2023
Postdoctoral Fellow Harvard University, <i>Cambridge, MA</i> Department of Earth and Planetary Sciences Note: disruptions in employment due to covid-19 and U.S. visa processing	Jul 2020 - Dec 2023
Graduate Research and Teaching Assistant Colorado State University, <i>Fort Collins, CO</i> Department of Atmospheric Sciences	Aug 2015 - May 2020
Climate Dynamics Intern Colorado State University, <i>Fort Collins, CO</i>	Mar - Jul 2015
Atmospheric Chemistry Intern National Center for Atmospheric Research, <i>Boulder, CO</i>	Mar - Jul 2014

PUBLICATIONS

Published

7. **L. Rivoire**, J. Curbelo, M. Linz, Tracking jets streams as Lagrangian objects, *Communications Earth & Environment*, 2026
6. **L. Rivoire**, M. Linz, P. Lin, J.N. Neu, M.L. Santee, Satellite nadir-viewing geometry affects the magnitude and detectability of long-term trends in stratospheric ozone, *Atmos Chem and Phys*, 2025.
5. H. Garny and co-authors including **L. Rivoire**, Age of stratospheric air: Progress on processes, observations and long-term trends, *Reviews of Geophysics*, 2024.
4. **L. Rivoire**, M. Linz, J. Li, Observational limitations to the emergence of climate signals, *Geophys. Res. Lett.*, 2024.
3. **L. Rivoire**, T. Birner, J. A. Knaff, N. Tourville, Quantifying the radiative impact of clouds on tropopause layer cooling in tropical cyclones, *Journal of Climate*, 2020.
2. W. J. Randel, **L. Rivoire**, L. L. Pan, S. Honomichl, Dry layers in the tropical troposphere observed during CONTRAST and global behavior from GFS analyses, *J. Geophys. Res.*, 2016.
1. **L. Rivoire**, T. Birner, J. A. Knaff, Evolution of the upper-level thermal structure in tropical cyclones, *Geophys. Res. Lett.*, 2016.

In review/preparation

L. Rivoire, S. Eastham, A. Fiore, J. Curbelo, Connectivity clusters as dynamical constraints on pollutant mixing and aging.

AWARDS, HONORS, AND GRANTS

Visiting Scientist Grant

Sep - Oct 2023

International Programme for Research in Groups, Center for Mathematical Research
Universitat Autònoma de Barcelona, *Barcelona, Spain*

Travel Grant

2019

19th Cyclone Workshop, *Kloster Seeon, Germany*

Programs of Research and Scholarly Excellence Award

2015

Colorado State University, *Fort Collins, CO*

TEACHING AND MENTORING

Research Mentoring Training Certificate

2024

Massachusetts Institute of Technology, *Cambridge, MA*

Leadership and Professional Strategies & Skills Training Certificate

2024

Massachusetts Institute of Technology, *Cambridge, MA*

Student Research Advisor

2022

Harvard College, *Cambridge, MA*

Teaching Assistant, Atmospheric Dynamics II

2016

Colorado State University, *Fort Collins, CO*

Student Intern Advisor

2016, 2019

Colorado State University, *Fort Collins, CO*

CONFERENCE PRESENTATIONS

Orals

American Meteorological Society Annual Meeting	2022, 2025 (2)
European Geosciences Union General Assembly, <i>Vienna, Austria</i>	2022, 2023, 2024, 2025, 2026
American Meteorological Society Conference on Hurricanes and Tropical Meteorology	2016, 2021
19th Cyclone Workshop, <i>Kloster Seeon, Germany</i>	2019
Front Range Tropical Cyclone Workshop, <i>Fort Collins, CO</i>	2018
NOAA Center for Satellite Applications and Research Symposium, <i>Madison, WI</i>	2018

Posters

American Geophysical Union Fall Meeting	2016, 2022, 2024
Stratospheric Processes And their Role in the Climate (SPARC) General Assembly, <i>Boulder, CO</i>	2022
American Meteorological Society Conf. on Hurricanes and Tropical Meteo., <i>Ponte Vedra, FL</i>	2018
Young Scientist Symposium on Atmospheric Research, <i>Fort Collins, CO</i>	2015

INVITED LECTURES AND SEMINARS

Universitat Autònoma de Barcelona, Center for Mathematical Research, <i>Barcelona, Spain</i>	2023
French and German Consulates, <i>Boston, MA</i>	2021
Colorado State University, Department of Atmospheric Science, <i>Fort Collins, CO</i>	2021
National Center for Atmospheric Research, <i>Boulder, CO</i>	2019
Colorado State University, Department of Atmospheric Science, <i>Fort Collins, CO</i>	2016

SERVICE AND LEADERSHIP

Peer review

Research Opportunities in Space and Earth Science (ROSES)	2017, 2019
NASA Headquarters, <i>Washington, D.C.</i>	

Harvard College Research Program	2022, 2023, 2024, 2025, 2026
Harvard University, <i>Cambridge, MA</i>	

Journal reviews ([Publons](#)):

Nature Reviews Earth & Env	Nature Communications	J Geophys Res	Bull Am Meteorol Soc
Atmos Sci Lett	Q J R Meteorol Soc	Atmos Chem Phys	Int J Climatol
Ann Geophys	Marine Pollution Bulletin	Int J Remote Sens	Earth & Space Science

Professional meetings

Session Lead Convener	
European Geosciences Union General Assembly, <i>Vienna, Austria</i>	2023, 2024, 2025, 2026

Student Poster Award Judge	
European Geosciences Union General Assembly, <i>Vienna, Austria</i>	2023, 2025
American Meteorological Society Annual Meeting, <i>New Orleans, LA</i>	2024